LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
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1.TITLE

Design Static Webpage Using HTML Components

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2. AIM

To design and develop a static webpage using core HTML components such as semantic tags, table-based layout, and native form elements, without using any CSS, and to understand how browsers render pure HTML structure by default.

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3. OBJECTIVE

- To understand the semantic structure of an HTML5 document.
- To use semantic tags such as <header>, <nav>, <main>, <article>, <section>, <aside>, and <footer>.
- To structure a webpage layout using HTML tables (rows, columns, cell padding/spacing).
- To implement HTML forms using various native input types (text, email, select, textarea, submit, reset).
- To group and label form controls using <fieldset> and <legend>.
- To display supplementary content using ordered and unordered lists.
- To observe how a browser renders a webpage with zero CSS styling.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to define the structure and content of a web page. Semantic HTML tags such as <header>, <nav>, <main>, <article>, <section>, <aside>, and <footer> describe the meaning and role of the content they enclose, rather than just its visual presentation. This improves accessibility for screen readers and helps browsers and search engines parse the document meaningfully.

HTML tables (<table>, <tr>, <td>) can also be used to arrange content into rows and columns for structural layout purposes, such as placing a title and navigation menu side by side, or dividing a page into a main content column and a sidebar column.

HTML forms (<form>, <input>, <select>, <textarea>, <fieldset>, <legend>) allow a webpage to collect user input directly through the browser's built-in controls. Attributes such as required and type="email" enable native client-side validation without any JavaScript.

Since no CSS is used anywhere in the document, the browser's default user-agent stylesheet handles all visual presentation, and the structure of the page is conveyed purely by which HTML tag encloses which piece of content.

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5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare <!DOCTYPE html> at the top.

2. Define the <head> section with <meta charset="UTF-8"> and a <title>.

3. Create a <header> section and use an HTML table to place the site title and navigation links side by side.

4. Insert a horizontal rule (<hr>) to visually separate the header from the next section.

5. Create a hero/showcase <section> with a heading and introductory text, centered using <center>.

6. Build the main content area using a two-column HTML table (70% / 30% widths).

7. In the left column, add a <main> element containing multiple <article> blocks, each with a heading, publish date, and paragraphs.

8. Inside an article, add a <form> containing a <fieldset> and <legend>, with text, email, select, and textarea inputs, plus submit and reset buttons.

9. In the right column, add an <aside> element with supplementary information using <ul> and <ol> lists.

10. Add a <footer> section containing copyright text.

11. Save the file with exp1.html extension.

12. Open the file in a web browser to verify that it renders correctly.

13. Test the form fields, dropdown, and validation behaviour.

14. Record observations and execution results as test cases.

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6. PROGRAM

Exp1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

   <!-- Header Section -->
   <header>
      <table width="100%" border="0" cellspacing="0" cellpadding="10">
         <tr>
            <td>
               <h1>StaticWeb</h1>
            </td>
            <td align="right">
               <nav>
                  <a href="#home">Home</a>  | 
                  <a href="#articles">Articles</a>  | 
                  <a href="#about">About</a>  | 
                  <a href="#contact">Contact Us</a>
               </nav>
            </td>
         </tr>
      </table>
   </header>

   <hr>
```

```html
<!-- Hero / Showcase Section -->
<section id="home">
   <center>
      <h2>Semantic Structures Without CSS</h2>
      <p>This layout uses raw HTML components to build an organized document hierarchy
that browsers can cleanly parse out-of-the-box.</p>
      <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
   </center>
</section>

<hr>

<!-- Main Content Layout using a Structural Table -->
<table width="100%" border="0" cellspacing="0" cellpadding="20">
   <tr valign="top">

      <!-- Left Side: Main Content Column -->
      <td width="70%">
         <main id="articles">

            <article>
               <h2>1. Core Structural Tags</h2>
               <p>Published: July 7, 2026</p>
               <p>When you omit cascading style sheets entirely, the semantic markup handles
all of the structural heavy lifting. Elements like <code>&lt;main&gt;</code>,
<code>&lt;article&gt;</code>, and <code>&lt;section&gt;</code> tell screen readers and web
scrapers exactly how information flows.</p>
               <p>Using raw browser defaults ensures excellent loading performance, perfect
accessibility compliance, and absolute structural clarity.</p>
            </article>

            <br><hr><br>

            <article>
               <h2>2. Native Interaction Elements</h2>
               <p>Published: July 5, 2026</p>
               <p>Browsers come equipped with excellent interactive elements by default. We
can capture text inputs, provide multiple-choice configurations, and submit structured
information to backend parsers using pure markup.</p>

               <h3>Interactive Demonstration Forms</h3>
               <form id="contact" action="#" method="post">
                  <fieldset>
                     <legend><b>User Inquiry Form</b></legend>
                     <p>
```

```
                        <label for="name">Full Name: </label>
                        <input type="text" id="name" name="username" required
placeholder="John Doe">
                    </p>
                    <p>
                        <label for="email">Email Address: </label>
                        <input type="email" id="email" name="useremail" required>
                    </p>
                    <p>
                        <label for="topic">Inquiry Topic: </label>
                        <select id="topic" name="usertopic">
                            <option value="general">General Feedback</option>
                            <option value="technical">Technical Architecture</option>
                            <option value="academics">Academic Curriculum</option>
                        </select>
                    </p>
                    <p>
                        <label for="message">Message Details:</label><br>
                        <textarea id="message" name="usermessage" rows="5" cols="40"
placeholder="Type your notes here..."></textarea>
                    </p>
                    <p>
                        <input type="submit" value="Submit Form Data">
                        <input type="reset" value="Clear Entries">
                    </p>
                </fieldset>
            </form>
        </article>

    </main>
</td>

<!-- Right Side: Sidebar Column -->
<td width="30%" bgcolor="#f4f4f4">
    <aside id="about">
        <h3>Document References</h3>
        <p>Static pages map data out directly without real-time database queries or
processor overhead.</p>

        <h4>Core Benefits:</h4>
        <ul>
            <li>Instant browser painting</li>
            <li>Low memory footprints</li>
            <li>High data compatibility</li>
        </ul>
```

```
                <h4>Required Components:</h4>
                <ol>
                    <li>Document Type Definition</li>
                    <li>Semantic Wrapper Blocks</li>
                    <li>Data Input Control Interfaces</li>
                </ol>
            </aside>
        </td>

    </tr>
</table>

<hr>

<!-- Footer Section -->
<footer>
    <center>
        <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style configurations.</p>
    </center>
</footer>

</body>
</html>
```

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7. CODE EXPLANATION

Tag / Element	**Purpose**
<!DOCTYPE html>	Declares the document as HTML5 so the browser uses standards mode.
<header>	Contains the site title and navigation bar at the top of the page.
<table> (in header)	Arranges the title and navigation links side by side without CSS.
<nav>	Wraps the group of navigation links (Home, Articles, About, Contact Us).
<section id="home">	Defines the hero/showcase area introducing the page content.
<center>	Centers the hero heading, paragraph, and link horizontally.
<table> (main layout)	Creates the two-column page layout (70% main content, 30% sidebar) using <tr> and <td>.
<main	Marks the primary, unique content area of the page (the articles list).

id="articles">	
<article>	Represents a single self-contained piece of content with its own heading and text.
<code>	Displays inline HTML tag names in a monospaced, code-like format.
<form id="contact">	Wraps all input controls; action="#" and method="post" define where/how data would be sent.
<fieldset> / <legend>	Groups related form controls together under the label "User Inquiry Form".
<input type="text">	Collects the user's full name; required forces the browser to validate it isn't empty.
<input type="email">	Collects an email address; the browser validates the format automatically.
<select> / <option>	Provides a dropdown to choose an inquiry topic (General/Technical/Academic).
<textarea>	Provides a multi-line box for the user's message details.
<input type="submit">	Submits the form data to the action URL.
<input type="reset">	Clears all entered form data back to its default state.
<aside id="about">	Sidebar holding supplementary content related to the main articles.
/	Display the core benefits (unordered) and required components (ordered) as lists.
<footer>	Contains the closing copyright notice, centered at the bottom of the page.

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8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open index.html in browser	Page loads with header, nav, hero section, articles, sidebar, footer	Page loaded correctly. Title "StaticWeb" displayed with nav menu, hero section, both articles,	Pass

			sidebar and footer all visible	
TC-02	Fill Name, Email, Topic, Message and click Submit	Form accepts data and submits	Form submitted successfully. Browser moved to index.html# (no error shown)	Pass
TC-03	Leave "Full Name" empty and click Submit	Browser shows required field error	Popup appeared under Name field: "Please fill out this field." Form did not submit	Pass
TC-04	Enter invalid email "abc" and click Submit	Browser shows invalid email error	Popup appeared under Email field: "Please include an '@' in the email address. 'abc' is missing an '@'." Form did not submit	Pass
TC-05	Fill all fields, then click "Clear Entries"	All fields reset to empty/default	Name field emptied, Email field emptied, Message box emptied, Topic dropdown went back to "General Feedback"	Pass

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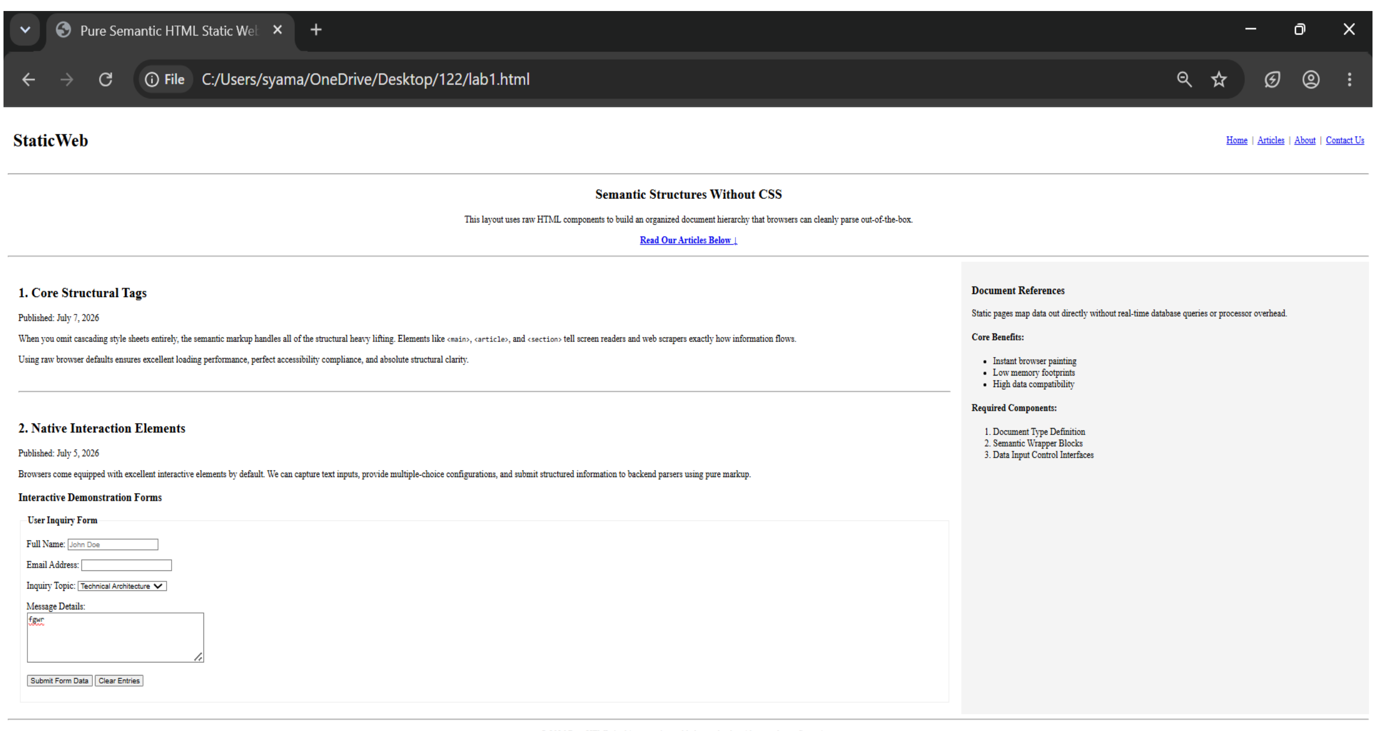
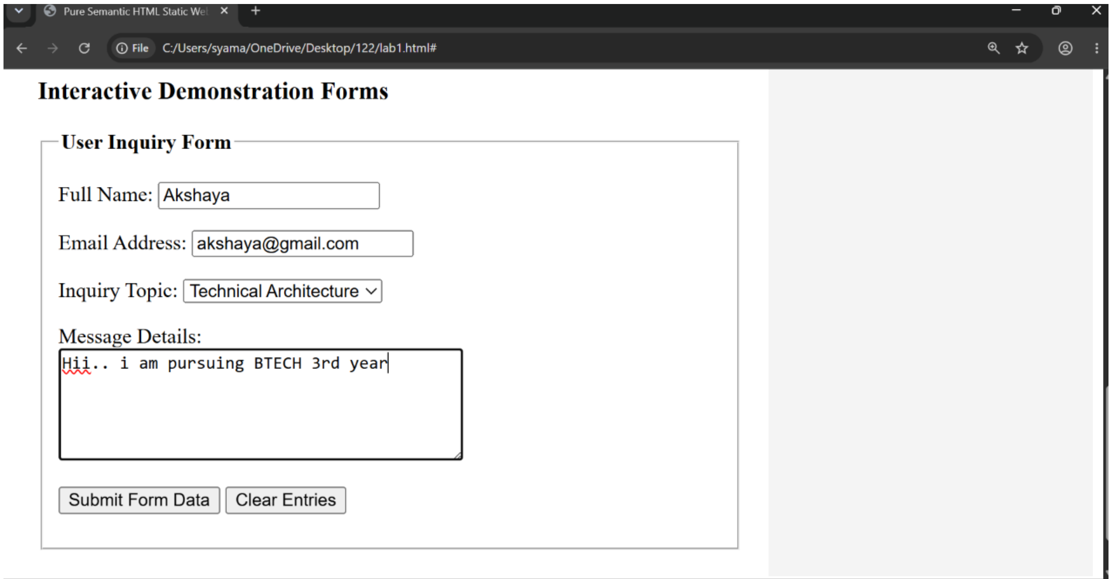
9. OUTPUT

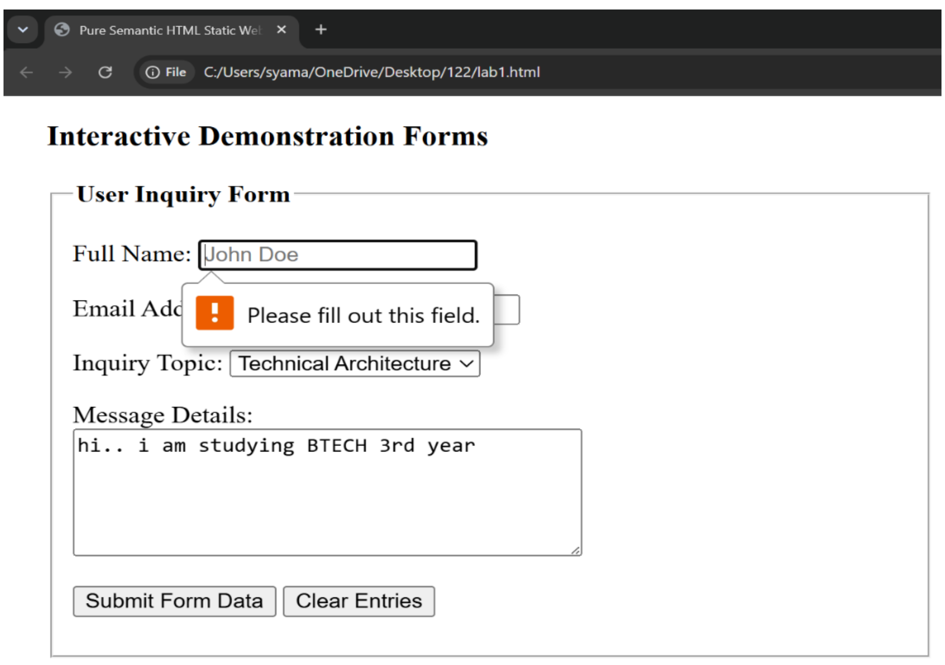
Output 1: Static Webpage



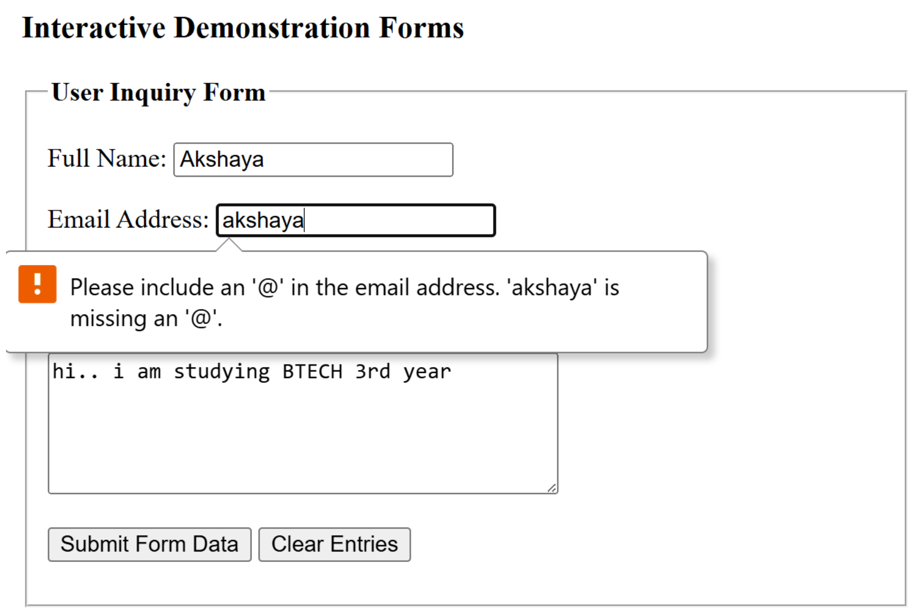
Output 2: Submitted Form Data



Output 3:



Output 4:



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10. RESULT

Thus, a static webpage was successfully designed and developed using pure HTML components, including semantic structural tags, table-based page layout, and native HTML form elements, without using any CSS. The webpage was executed and tested in a web browser, and all test cases — covering page rendering, form submission, required-field validation, email-format validation, and form reset — were verified and found to work correctly.